Connection of Java with MySQL Database

Aim :- To connect Java with MySQL database and perform operations such as insert, view, and navigate student records.

Problem Statement :- Write a Java program to establish a connection between Java and MySQL database using JDBC and insert student records into the database.

Software and Tools Used :-

- Programming Language: Java
- Database: MySQL
- IDE: IntelliJ
- JDBC Driver: MySQL Connector/J

SQL Studentdb database created :-

```
create database studentdb;

use studentdb;

create table student(rno int,name varchar(50),course
varchar(20),fees float);

insert into student values(1,'shreya','java',5000);

select * from student;
```

Java program:-

```
package pack_studentDB;
```

```
import javax.swing.*;
import java.awt.*;
import java.awt.event.*;
import java.sql.*;

public class StudentGui implements ActionListener {
   JFrame jFrame;
   JLabel jLabelrno, jLabelnm, jLabelcourse, jLabelfee;
   JTextField jTextFieldrno, jTextFieldname, jTextFieldcourse,
jTextFieldfee;
   JButton jButtonfirst, jButtonnext, jButtonprev, jButtonlast,
jButtonnew, jButtoninsert, jButtonupdte, jButtondelete, jButtonclear,
jButtonexit;
   Font font;

   Connection con;
   PreparedStatement pstmt;
   ResultSet rs;

   public StudentGui() {
      font = new Font("Arial", Font.BOLD, 16);
      jFrame = new JFrame("Student Database");
      jFrame.setBounds(100, 100, 500, 450);
      jFrame.setLayout(null);
      jFrame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);

      // Labels
      jLabelrno = new JLabel("Roll No:");
      jLabelrno.setBounds(50, 40, 100, 30);
      jLabelrno.setFont(font);
      jFrame.add(jLabelrno);

      jLabelnm = new JLabel("Name:");
```

```java
jLabelnm.setBounds(50, 80, 100, 30);
jLabelnm.setFont(font);
jFrame.add(jLabelnm);

jLabelcourse = new JLabel("Course:");
jLabelcourse.setBounds(50, 120, 100, 30);
jLabelcourse.setFont(font);
jFrame.add(jLabelcourse);

jLabelfee = new JLabel("Fees:");
jLabelfee.setBounds(50, 160, 100, 30);
jLabelfee.setFont(font);
jFrame.add(jLabelfee);

// TextFields
jTextFieldrno = new JTextField();
jTextFieldrno.setBounds(160, 40, 200, 30);
jTextFieldrno.setFont(font);
jFrame.add(jTextFieldrno);

jTextFieldname = new JTextField();
jTextFieldname.setBounds(160, 80, 200, 30);
jTextFieldname.setFont(font);
jFrame.add(jTextFieldname);

jTextFieldcourse = new JTextField();
jTextFieldcourse.setBounds(160, 120, 200, 30);
jTextFieldcourse.setFont(font);
jFrame.add(jTextFieldcourse);

jTextFieldfee = new JTextField();
jTextFieldfee.setBounds(160, 160, 200, 30);
jTextFieldfee.setFont(font);
```

```java
        jFrame.add(jTextFieldfee);

        // Buttons
        jButtonfirst = new JButton("First");
        jButtonnext = new JButton("Next");
        jButtonprev = new JButton("Prev");
        jButtonlast = new JButton("Last");
        jButtonnew = new JButton("New");
        jButtoninsert = new JButton("Insert");
        jButtonupdte = new JButton("Update");
        jButtondelete = new JButton("Delete");
        jButtonclear = new JButton("Clear");
        jButtonexit = new JButton("Exit");

        JButton[] btns = {jButtonfirst, jButtonnext, jButtonprev,
jButtonlast, jButtonnew,
                jButtoninsert, jButtonupdte, jButtondelete, jButtonclear,
jButtonexit};

        int x = 20, y = 220;
        for (int i = 0; i < btns.length; i++) {
            btns[i].setBounds(x, y, 100, 35);
            btns[i].setFont(font);
            btns[i].addActionListener(this);
            jFrame.add(btns[i]);
            x += 110;
            if (x > 400) {
                x = 20;
                y += 50;
            }
        }

        jFrame.setVisible(true);
```

```
        // Connect to Database
        connectToDB();
    }

    public void connectToDB() {
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            con = DriverManager.getConnection(

"jdbc:mysql://localhost:3306/studentdb?useSSL=false&serverTimezo
ne=UTC",
                "root",
                "sa123" // change to your password
            );
            getDataFromDB();
        } catch (Exception ex) {
            JOptionPane.showMessageDialog(jFrame, "Database
Connection Failed: " + ex.getMessage());
        }
    }

    public void getDataFromDB() {
        try {
            pstmt = con.prepareStatement("SELECT * FROM student",
                ResultSet.TYPE_SCROLL_INSENSITIVE,
                ResultSet.CONCUR_READ_ONLY);
            rs = pstmt.executeQuery();
            if (rs.next()) {
                displayData();
            }
        } catch (Exception ex) {
            ex.printStackTrace();
```

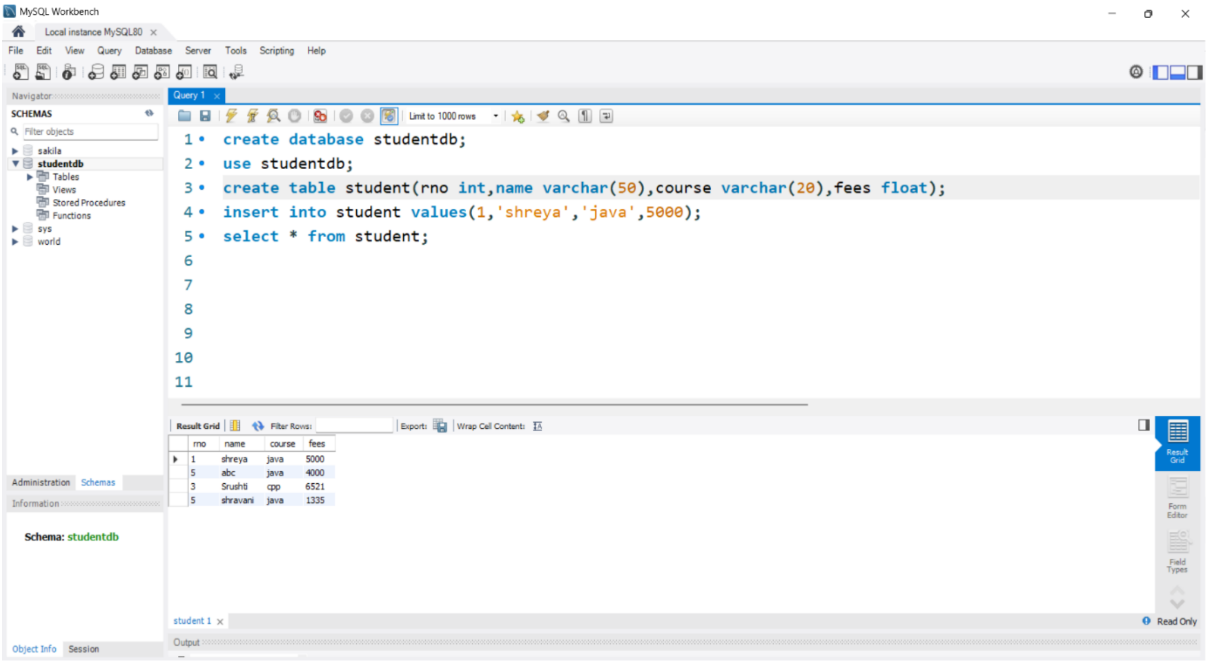
```
        }
    }

    public void displayData() {
        try {
            jTextFieldrno.setText(rs.getString("rno"));
            jTextFieldname.setText(rs.getString("name"));
            jTextFieldcourse.setText(rs.getString("course"));
            jTextFieldfee.setText(rs.getString("fees"));
        } catch (Exception ex) {
            ex.printStackTrace();
        }
    }

    public void insertDataIntoDB() {
        try {
            int rno = Integer.parseInt(jTextFieldrno.getText());
            String name = jTextFieldname.getText();
            String course = jTextFieldcourse.getText();
            float fee = Float.parseFloat(jTextFieldfee.getText());

            String sql = "INSERT INTO student VALUES(?, ?, ?, ?)";
            pstmt = con.prepareStatement(sql);
            pstmt.setInt(1, rno);
            pstmt.setString(2, name);
            pstmt.setString(3, course);
            pstmt.setFloat(4, fee);

            int n = pstmt.executeUpdate();
            if (n > 0) {
                JOptionPane.showMessageDialog(jFrame, "Record Inserted
Successfully!");
                getDataFromDB();
```

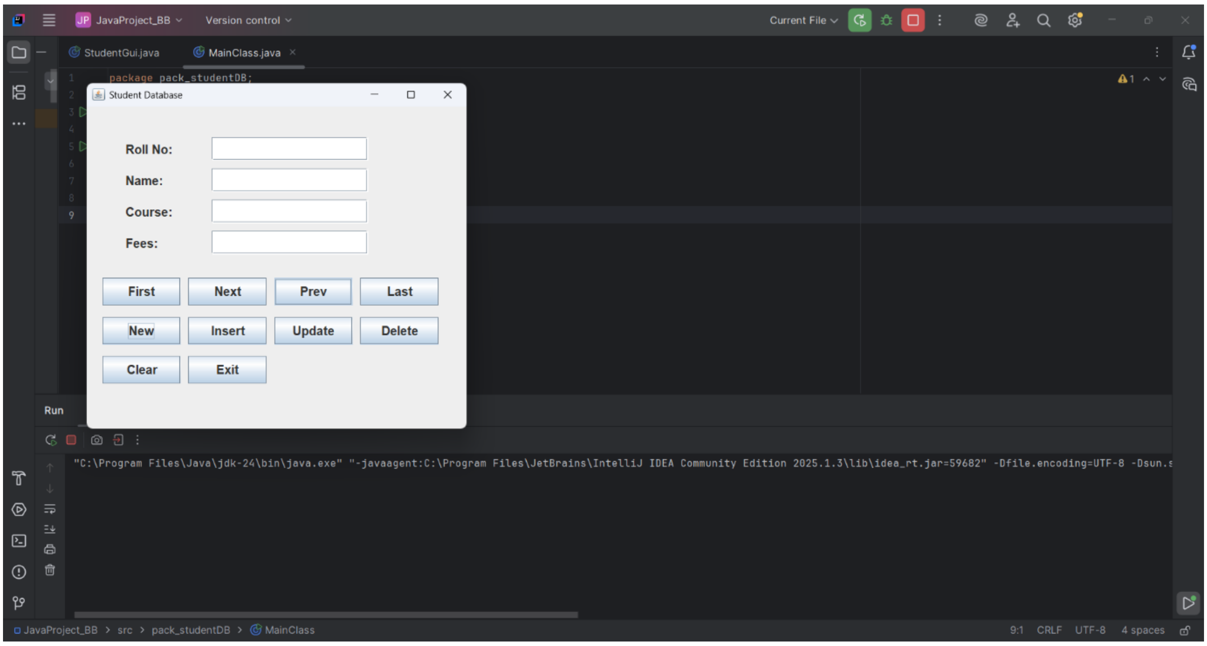
```
        }
      } catch (Exception e) {
        JOptionPane.showMessageDialog(jFrame, "Error: " +
e.getMessage());
      }
    }

    public void clearALL() {
      jTextFieldrno.setText("");
      jTextFieldname.setText("");
      jTextFieldcourse.setText("");
      jTextFieldfee.setText("");
    }

    public void actionPerformed(ActionEvent e) {
      try {
        Object src = e.getSource();

        if (src == jButtonfirst) {
          if (rs.first()) displayData();
        } else if (src == jButtonnext) {
          if (rs.next()) displayData();
          else JOptionPane.showMessageDialog(jFrame, "Last
Record");
        } else if (src == jButtonprev) {
          if (rs.previous()) displayData();
          else JOptionPane.showMessageDialog(jFrame, "First
Record");
        } else if (src == jButtonlast) {
          if (rs.last()) displayData();
        } else if (src == jButtonnew || src == jButtonclear) {
          clearALL();
        } else if (src == jButtoninsert) {
```

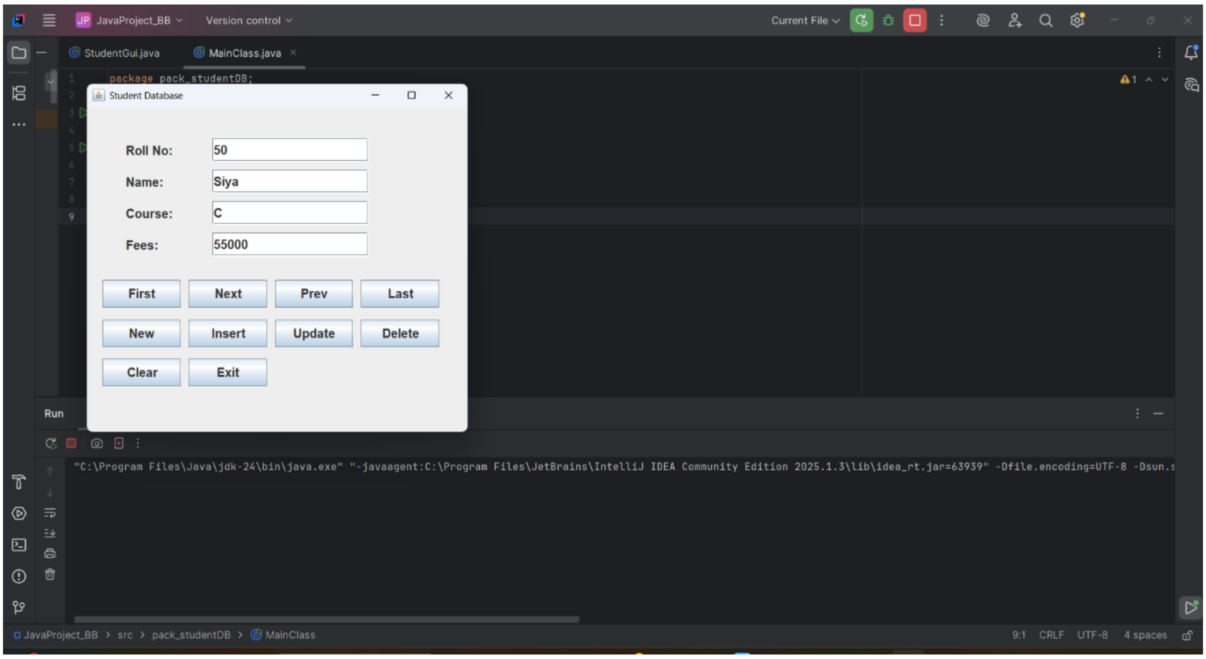
```
            insertDataIntoDB();
        } else if (src == jButtonexit) {
            System.exit(0);
        }

    } catch (Exception ex) {
        ex.printStackTrace();
    }
}

public static void main(String[] args) {
    new StudentGui();
}
}
```

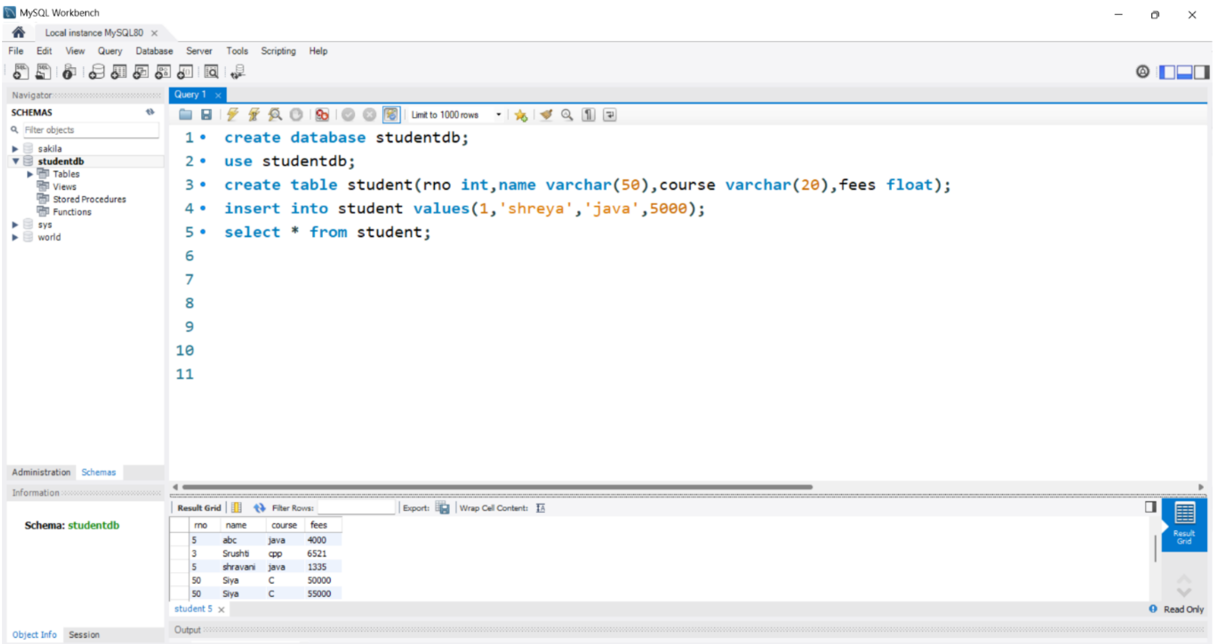
```
package pack_studentDB;

public class MainClass
{
    public static void main(String[] args) {
        StudentGui gui = new StudentGui();
    }
}
```

Output:-


MySQL Workbench
Local instance MySQL80
File Edit View Query Database Server Tools Scripting Help
Query 1
create database studentdb;
use studentdb;
create table student(rno int,name varchar(50),course varchar(20),fees float);
insert into student values(1,'shreya','java',5000);
select * from student;
rno name course fees
1 shreya java 5000
5 abc java 4000
3 Srushti cpp 6521
5 shravani java 1335
Schema: studentdb


JavaProject_BB
StudentGui.java
MainClass.java
package pack_studentDB;
Student Database
Roll No:
Name:
Course:
Fees:
First
Next
Prev
Last
New
Insert
Update
Delete
Clear
Exit
Run
"C:\Program Files\Java\jdk-24\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA Community Edition 2025.1.3\lib\idea_rt.jar=59682" -Dfile.encoding=UTF-8
JavaProject_BB > src > pack_studentDB > MainClass





A GUI window displays Roll No, Name, Course, and Fees with navigation and insert options. Database successfully connected and records inserted into MySQL.

Conclusion :-

This project demonstrates Java Database Connectivity (JDBC) — a bridge between Java and relational databases. It helps in performing CRUD (Create, Read, Update, Delete) operations on databases directly through a Java program.